

In the previous exercise, we note that the result was based on knowledge of the distribution of the test statistic under the null hypothesis. It is key that the test statistic be chosen so that its distribution is known (at least approximately), under the null. As usual, the central limit theorem can be very helpful.

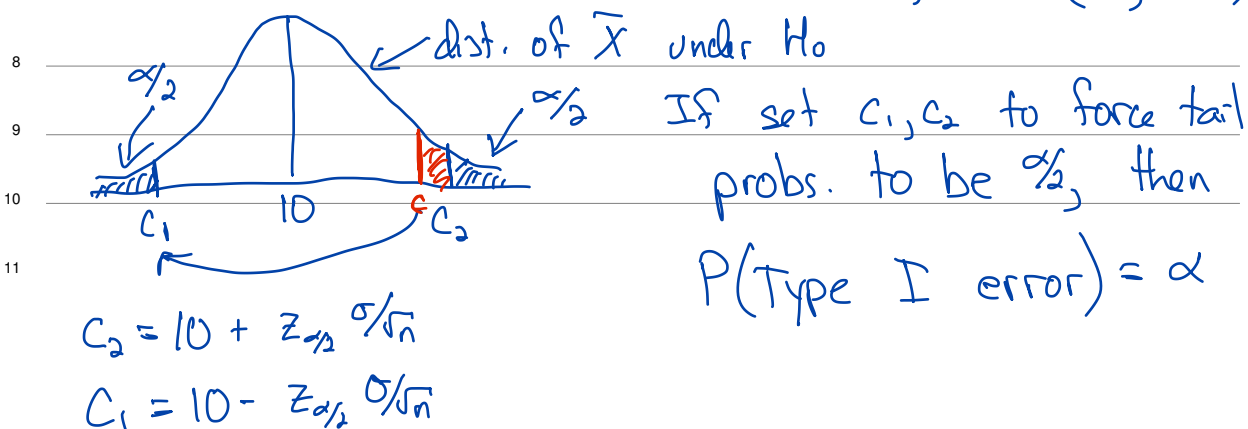
Also, we note that in the exercise we tested the one-sided alternative hypothesis $H_1: \mu > 10$. In other cases we will test a two-sided alternative, for example, $H_1: \mu \neq 10$. In such a case the null would be rejected for either small or large values of the test statistic.

$$H_0: p = 0.5 \quad \text{vs.} \quad H_1: p \neq 0.5$$

Exercise: Suppose that we tested $H_1: \mu \neq 10$ in the above example. What would the rejection region be for fixed α ?

Now reject $H_0: \mu = 10$ if $\bar{X}$ is sufficiently smaller or larger than 10, i.e. RR is of the form $(-\infty, c_1) \cup (c_2, \infty)$ where $c_1 < 10 < c_2$

Think about dist. of $\bar{X}$ under H_0 , i.e. $N(10, \sigma^2/n)$



1 The p-value

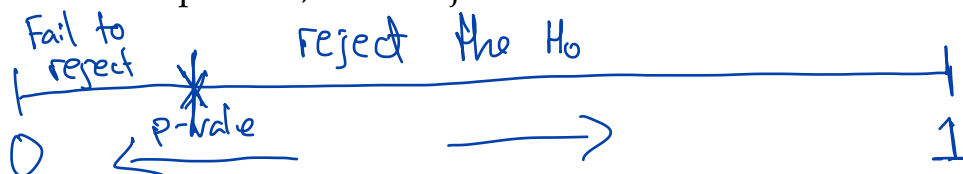
2 The choice of α in the statistical hypothesis testing framework may seem
3 arbitrary. Why would one choose one value of α over another?

4 As α is chosen smaller, the experimenter is choosing to be more cautious,
5 to protect against Type I error. The opposite is true as α is chosen larger.

6 For a given data set, there is a **dividing line** between the choices of α for
7 which the conclusion is “reject the null” and the conclusion is “fail to reject
8 the null.” This dividing line is the **p-value**.

9 1. If $\alpha < \text{p-value}$, then “fail to reject the null.”

10 2. If $\alpha > \text{p-value}$, then “reject the null.”



1 It is important to note that the p-value is **a function of the data**.

2 The p-value is **not** the probability that the null hypothesis is true. Instead,
3 it is the largest choice of α such that the conclusion is “fail to reject the
4 null.”

5 The appeal is that one can report a p-value, and allow the reader to decide.

6 Most software packages report the results of hypothesis tests as p-values.

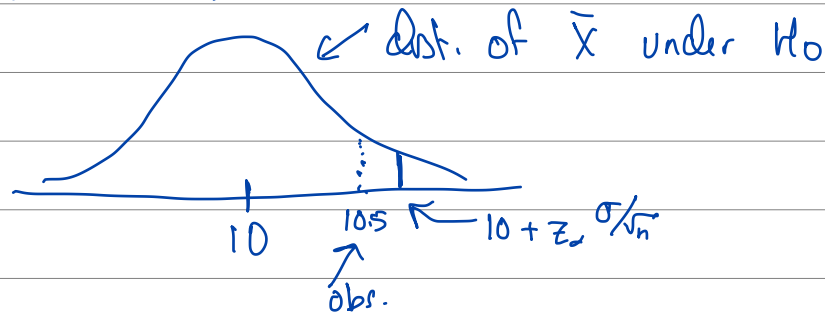
7 The next exercise illustrates another interpretation of the p-value: It is
8 the probability that, assuming the null hypothesis is true, the test statis-
9 tic would be equal to or more extreme in favor of the alternative than the
10 observed test statistic.

11 This interpretation is often useful for making computations.

$$\begin{array}{cccc} \hat{\beta}_1 & \hat{SE} & \hat{t} & p \\ \hline & & & \\ & & \nearrow & \nearrow \\ & & H_0: \beta_1 = 0 & \\ & & \text{vs. } H_1: \beta_1 \neq 0 & \end{array}$$

- 1 **Exercise:** Return to our example when testing $H_0: \mu = 10$ versus $H_1: \mu >$
 2 10 . Assume that $\sigma^2 = 1$. We observe $\bar{x} = 10.5$ with a sample size of $n = 50$.
 3 What is the p-value in this case? What happens if we switch to $H_1: \mu \neq 10$?

4 When testing $H_0: \mu = 10$ vs. $H_1: \mu > 10$ we know
 5 reject H_0 if $\bar{x} > 10 + z_\alpha \sigma/\sqrt{n}$



11 When would we reject?

12 $10 + z_\alpha \sigma/\sqrt{n} < 10.5 \swarrow$

1 We would fail to reject if
 2 $10 + z_\alpha \sigma/\sqrt{n} > 10.5 \swarrow$

3 What choice of α puts us on the dividing line?

4 When $10 + z_\alpha \sigma/\sqrt{n} = 10.5$

5
6
7 ie. when $z_\alpha = (0.5) \left(\frac{\sqrt{n}}{\sigma} \right) = (0.5) \left(\frac{\sqrt{50}}{1} \right) = 3.53$

8 What choice of α forces $z_\alpha = 3.53$?

9 $\alpha = P(Z > 3.53) = 0.000208$

10 So, p-value = 0.000208

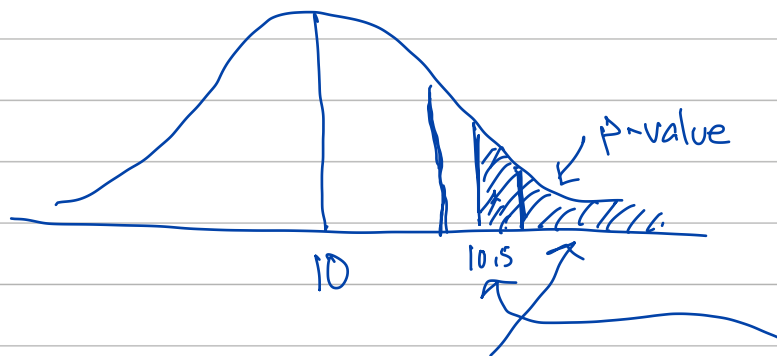


12 "Only people who choose $\alpha < 0.000208$ would fail to reject H_0 ."

Another way to think of it:

What choice of α forces the cutoff to be 10.5?
 $\nwarrow$ observed test stat.

It is exactly the prob. above 10.5 under the $N(10, \sigma/\sqrt{n})$ dist.



$$P(\text{test stat} \geq \text{Obs. test stat} \mid H_0 \text{ true})$$

In two-sided case: $H_0: \mu = 10$ vs. $H_1: \mu \neq 10$. Observe $\bar{X} = 10.5$

$$\begin{aligned} \text{p-value} &= P(\bar{X} \geq 10.5 \text{ or } \bar{X} \leq 9.5 \mid H_0 \text{ true}) \\ &\stackrel{\text{by symmetry}}{=} 2 P(\bar{X} \geq 10.5 \mid H_0 \text{ true}) \end{aligned}$$

1 The Wald Test

2 The **Wald Test** is built around a simple notion: An ideal choice for the test
 3 statistic is the number of standard errors a parameter estimate is from the
 4 null hypothesized value for that parameter.

5 Specifically, if testing $H_0: \theta = \theta_0$, then the test statistic is chosen as

$$T = \frac{\hat{\theta} - \theta_0}{\text{SE}(\hat{\theta})}$$

(signed)

7 T gives the "the number of standard errors $\hat{\theta}$ is from θ_0 ."

8 In a wide range of situations, T has approximately the standard normal
 9 distribution under the null hypothesis. Hence, the region region is not
 10 difficult to derive.

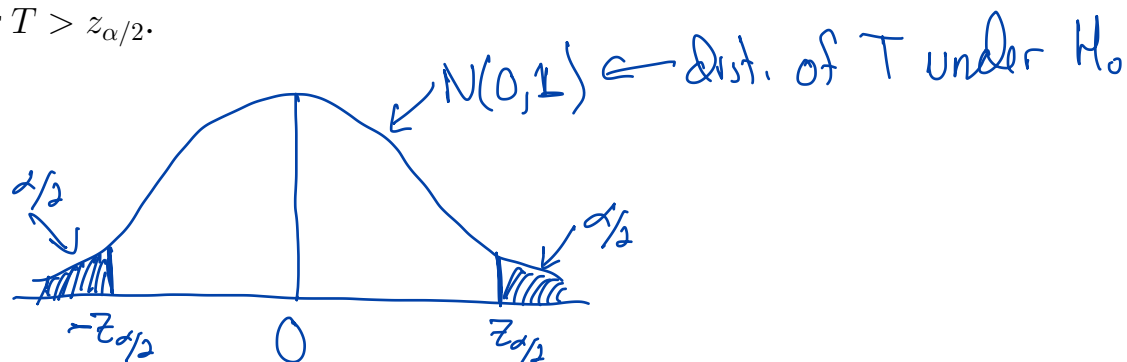
1 **Exercise:** What is one very important case in which we can state that T is
 2 approximately standard normal under the null?

$\hat{\theta}$ is the MLE for θ

$$\text{SE}(\hat{\theta}) = \frac{1}{\sqrt{nI(\theta_0)}}$$

$$T = \frac{\hat{\theta} - \theta_0}{\sqrt{\frac{1}{nI(\theta_0)}}} \text{ is approx. } N(0, 1) \text{ under } H_0$$

- The three possible tests can be presented as follows:
1. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta > \theta_0$, then reject the null if $T > z_\alpha$.
 2. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta < \theta_0$, then reject the null if $T < -z_\alpha$.
 3. If testing $H_0: \theta = \theta_0$ versus $H_1: \theta \neq \theta_0$, then reject the null if either $T < -z_{\alpha/2}$ or $T > z_{\alpha/2}$.



- Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. Bernoulli(p). Construct the two-sided Wald Test for p . Suppose that $n = 100$ and $\sum X_i = 47$. Test $H_0: p = 0.4$ versus $H_1: p \neq 0.4$. What is the p-value?

The MLE is $\hat{p} = \sum X_i / n$ (sample prop.)

$$I(p) = \frac{1}{p(1-p)}, \quad SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \leftarrow$$

So, the Wald Test stat for $H_0: p = p_0$

$$T = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

1 We want to test $H_1: p \neq p_0$, so reject H_0
 2 if $T > z_{\alpha/2}$ or $T < -z_{\alpha/2}$

3 In example, $\sum X_i = 47$, $n=100$ $\hat{p} = 0.47$
 4 $p_0 = 0.40$

$$5 \quad T = \frac{0.47 - 0.4}{\sqrt{\frac{(0.4)(0.6)}{100}}} = 1.43$$

6 If I choose $\alpha = 0.05$, $z_{\alpha/2} = 1.96$, so fail to reject
 7 " " " $\alpha = 0.10$, $z_{\alpha/2} = 1.645$, " " " "
 8

9 The p-value is larger than 0.10
 10
 11
 12

1 The exact p-value:

$$2 \quad P(T \geq 1.43 \text{ or } T \leq -1.43 \mid H_0 \text{ true})$$

$$3 \quad = 2P(Z \geq 1.43) \quad Z \sim N(0,1)$$

$$4 \quad \approx 0.15$$

5 Note: If $H_1: p > 0.4$, then p-value ≈ 0.075
 6
 7
 8
 9
 10

- 1 The Wald Test can be extended to testing multiple parameters simultane-
 2 ously by noting the following result: If $\mathbf{X}$ is p -dimensional multivariate
 3 normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, then

4
$$(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu})$$
 ←

- 5 has the chi-squared distribution with p degrees of freedom.

- 6 **Exercise:** Explain how this result can be used in conjunction with a multi-
 7 parameter distribution when the MLE is used as the estimator.

8 " $\mathbf{X}$ " replaced with $\hat{\boldsymbol{\theta}}_{MLE}$ " $\boldsymbol{\mu}$ " replaced with $\boldsymbol{\theta}_0$
 9 " $\boldsymbol{\Sigma}$ " replaced with $\mathbf{I}^{-1}(\boldsymbol{\theta}_0)/n$, $\boldsymbol{\Sigma}^{-1} = n\mathbf{I}^{-1}(\boldsymbol{\theta}_0)$

10
$$T = (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)^T \underline{n\mathbf{I}^{-1}(\boldsymbol{\theta}_0)} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$$

11 $T \sim \chi_p^2$ under $H_0: \boldsymbol{\theta} = \boldsymbol{\theta}_0$ ← a distance between $\hat{\boldsymbol{\theta}}$ and $\boldsymbol{\theta}_0$
 12 reject H_0 if T is large,

$$p\text{-value} = P(T \geq T_{obs} \mid T \sim \chi_p^2)$$

$$H_1: \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$$

Part 19: More Hypothesis Testing

The Multiple Testing Problem

Possibly the biggest drawback of the hypothesis testing framework is the **multiple testing problem**.

Exercise: Suppose a sequence of 100 independent hypothesis tests are conducted, in every case the null hypothesis is true, and $\alpha = 0.05$ is used. How many of these tests will reject the null?

$$\# \text{ of rejections} \sim \text{binomial}(n=100, p=0.05)$$

$$E(\# \text{ of rejections}) = 5$$

Exercise: Suppose that m independent hypothesis tests are conducted, in every case the null hypothesis is true. What is the probability of at least one rejection, if the same α is used for all tests?

$$\begin{aligned} P(\text{at least 1 rejection}) &= 1 - P(\text{no rejections}) \\ &= 1 - (1 - \alpha)^m \end{aligned}$$

$$\text{When } m=10, \alpha=0.05, P(\text{at least 1 rejection}) = 0.4 \gg \alpha$$

Assume remove independence.

$$P(\text{at least 1 rejection}) \text{ could be as large as } \max P(A_i \cup A_j \cup \dots \cup A_m) \leq \sum P(A_i) = m\alpha$$

$$\alpha_0 = 0.05 \Rightarrow \alpha = \alpha_0 / m$$

Use on each test

Leads to Bonferroni correction

In recognition of the multiple testing problem, recent advances have been made in controlling the **false discovery rate (FDR)**, instead of worrying about making one or more Type I error.

The false discovery rate can be defined as follows:

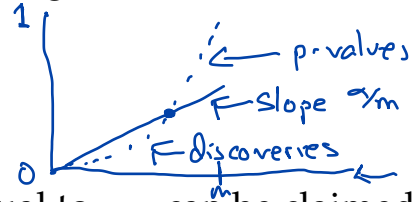
$$\text{FDR} = \frac{\text{number of false positives}}{\text{number of rejections}} \quad \# \text{ of discovery claims}$$

Hence, FDR is the proportion of the rejected null hypotheses which are "false discoveries."

We should be clear: This approach is "solving" the multiple testing problem by, more or less, changing the objective. Still, FDR seems like a very reasonable alternative to control instead of limiting the probability of a Type I error.

Benjamini and Hochberg (*JRSSB*, Vol. 57) presented a simple procedure which limits the expected value of the FDR below a chosen α level:

1. Sort the p-values from smallest to largest, denote these $p_{(1)}, p_{(2)}, \dots, p_{(m)}$.
2. Find the largest k such that $p_{(k)} \leq k\alpha/m$
3. The hypotheses with p-values less than or equal to $p_{(k)}$ can be claimed as “discoveries” for this choice of α .



It should be stressed that the validity of this procedure assumes that the individual hypothesis tests are independent or satisfy certain dependency conditions. Extensions do exist; for example, the **Benjamini-Hochberg-Yekutieli procedure** divides the factor $k\alpha/m$ by $\sum_{i=1}^m 1/i$ and works under a wider range of dependence situations.

The function `multipletests()` from `statsmodels`, when used with argument `method="fdr_bh"` will adjust the supplied p-values, returning the smallest choice of α such that a discovery would be claimed for each test. The choice `method="fdr_by"` implements the Benjamini-Hochberg-Yekutieli procedure.

The syntax is illustrated below. Note that the first two components returned from the function are (1) a boolean vector (indicating whether or not that p-value is small enough to claim a discovery) and (2) an “adjusted” p-value that can be interpreted as the smallest choice of α for which this test corresponds to a discovery.

```
In [1]: from statsmodels.stats.multitest \
        import multipletests
        from scipy.stats import uniform
        import numpy as np
```

```
simpvals = uniform.rvs(size=1000)
discovery, adjpval = multipletests(simpvals,
    alpha = 0.5, method="fdr_bh")[0:2]
```

Fact: If H_0 is true, $p\text{-value} \sim \text{Unif}(0,1)$

Choose test so that $P(\text{Type I error}) = \alpha$

i.e., $P(\text{reject } H_0 | H_0 \text{ true}) = \alpha$

$P(p\text{-value} \leq \alpha | H_0 \text{ true}) = \alpha \leftarrow$

Hence, under H_0 , $p\text{-value} \sim U(0,1)$

(If $U \sim \text{Unif}(0,1)$, $P(U \leq c) = c$ for $0 < c < 1$)

Exercise: Consider the output below and comment.

$\text{adjpval} < 0.05$ returns True if can
claim discovery at choice $\alpha = 0.05$
None of the 1000 tests meet that standard

Even if I choose $\alpha = 0.5$ only make one
discovery claim

Of course, we know that there are no "true"
discoveries here

```
In [2]: print(np.mean(simpvals < 0.05))
        print(np.mean(simpvals < 0.5))

        print(np.mean(adjpval < 0.05))
        print(np.mean(adjpval < 0.5))
```

← prop. of p-values < 0.05

←

0.058
0.515
0.0
0.001

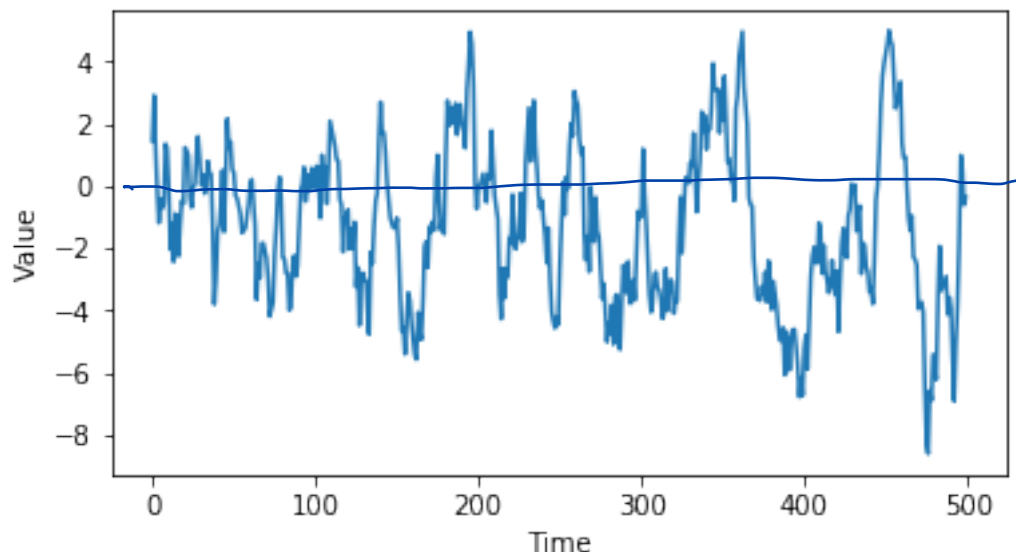
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Example: Simple Pairs Trading

A **time series** is a sequence of random variables (or, the actual observed values of those random variables) that model a sequence of data over time. Typically these observations are assumed to be equally spaced in time. Denote this $X_1, X_2, \dots$ or $\{X_t\}$.

A time series is said to be **stationary** if the **unconditional** mean and variance is constant ($E(X_i) = \mu$ and $V(X_i) = \sigma^2$) and the covariance structure also does not change with time, i.e., $\text{Cov}(X_t, X_{t+h})$ only depends on h , and not on t , for any choice of h and t . Stationarity is a crucial concept for time series models.

Stationary time series exhibit **mean reversion**: The series may move away from μ for periods of time, but it will inevitably drift back to μ . The example below shows a series simulated from the AR(1) model, with $\mu = 0$.



Exercise: Can time series of stock prices be modelled using stationary time series models?

No. Mean reversion would imply a simple way to make money.

A pair of time series $\{X_t\}$ and $\{Y_t\}$ are said to be **cointegrated** if there exists a constant λ such that $X_t - \lambda Y_t$ is stationary. Note that this does not require that either $\{X_t\}$ or $\{Y_t\}$ be stationary.

This sets up a simple **pairs trading** strategy: If one can find pairs of stocks whose prices are cointegrated, then we can use mean reversion of $X_t - \lambda Y_t$ to execute trades that will reliably make money.

Hypothesis tests do exist that test for cointegration. Here, we will utilize the **Phillips-Ouliaris Test** for assessing co-integration.

Exercise: What is the null hypothesis with this cointegration test? Does that seem to be the reasonable choice?

H_0 : no co-integrating relationship
 H_1 : Is cointegration

This is the appropriate. We want to determine if there is strong evidence for cointegration before we implement this trading strategy.

Exercise: One could search over many pairs of stocks in an attempt to find cointegrated pairs. What are potential problems with this approach?

Huge multiple testing problem.

With p series, $\binom{p}{2}$ tests.

Potential for many false rejections

I constructed a data set consisting of 1000 NYSE closing price time series, covering the period January through August 2020.

```
In [342]: import pandas as pd
          import numpy as np

          stocksample = pd.read_csv("stocksample2.csv")

          stockdat = stocksample.iloc[:,1:]
          syms = stocksample['Symbol']
          numsyms = len(syms)
          numtests = int(numsyms/2)
```

Next randomly pair up the symbols:

```
In [301]: np.random.seed(0)

          sp = np.array(np.random.choice(range(numsyms),
          size=[int(numtests),2],replace=False))
```

Now calculate the p-values for each of the tests:

```
In [343]: from arch.unitroot.cointegration \
          import phillips_ouliaris

          pval = np.zeros(numtests)

          for i in range(numtests):
              potest = phillips_ouliaris(
                  stockdat.iloc[sp[i][0],:],
                  stockdat.iloc[sp[i][1],:], trend="n")
              pval[i] = potest._pvalue
```

How many of the p-values are less than 0.05?

```
In [341]: sum(pval < 0.05)
```

```
Out [341]: 58
```

What happens when we try to control the false discovery rate?

```
In [338]: discovery, adjpval = multipletests(pval,  
      alpha = 0.05, method='fdr_by') [0:2]  
  
      sum(discovery)
```

```
Out [338]: 5
```

This is a good example of why one needs to be careful of the multiple testing problem. If you search over enough pairs, some are going to appear to be cointegrated. False positives are inevitable.

Note: p-values approx. using simulations, and may not be valid in the lower tail